

COMPLEX NUMBER

- **Q. Solve** $x^2 - 1 = 0$

Ans: $-x^2 - 1 = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$

- **Q. Solve** $x^2 + 1 = 0$

Ans: $-x^2 + 1 = 0 \Rightarrow x^2 = -1 \Rightarrow x = \pm\sqrt{-1}$. So there is no real solution for the given question.

As $\sqrt{-1}$ is an imaginary number denoted by i known as iota.

Hence $i = \sqrt{-1}$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$

In general $i^{4n+1} = \sqrt{-1}$, $i^{4n+2} = -1$, $i^{4n+3} = -i$, $i^{4n} = 1$ where $n \in N$.

- **Complex Number:** - A complex number is of the form $z = x + iy$ where $x, y \in R$.

Real part of $z = \text{Re}(z) = x$ and Imaginary part of $z = \text{Im}(z) = y$

Also complex number $z = x + iy$ can be denoted as $z(x, y)$.

- **Ordering of a Complex Number:** There is no ordering in complex numbers. We cannot say which one is greater and which one is smaller.
- **Equality of Two Complex Numbers:** - Two complex numbers are said to be equal if their corresponding real parts and corresponding imaginary parts are equal.
- **Purely Real Number:** - A complex number is said to be purely real number if it has no imaginary part or imaginary part is zero.
So, $z = x + i0 = x = (x, 0)$.
- **Purely Imaginary Number:** - A complex number is said to be purely imaginary number if it has no real part or real part is zero. So, $z = 0 + iy = iy = (0, y)$

- **Addition & Subtraction of Complex Numbers:** -

Let $z_1 = x_1 + iy_1 = (x_1, y_1)$ and $z_2 = x_2 + iy_2 = (x_2, y_2)$

Hence $z_1 + z_2 = x_1 + iy_1 + x_2 + iy_2 = (x_1 + x_2) + i(y_1 + y_2) = (x_1 + x_2, y_1 + y_2)$

Similarly $z_1 - z_2 = x_1 + iy_1 - x_2 - iy_2 = (x_1 - x_2) + i(y_1 - y_2) = (x_1 - x_2, y_1 - y_2)$

- **Multiplication of Two Complex Numbers:** -

Let $z_1 = x_1 + iy_1 = (x_1, y_1)$ and $z_2 = x_2 + iy_2 = (x_2, y_2)$

Hence $z_1 z_2 = (x_1 + iy_1)(x_2 + iy_2) = x_1(x_2 + iy_2) + iy_1(x_2 + iy_2)$

$\Rightarrow z_1 z_2 = x_1 x_2 + ix_1 y_2 + ix_2 y_1 - y_1 y_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1)$

- **Inverse of A Complex Number:** - Let $z = x + iy$

Then $z^{-1} = \frac{1}{z} = \frac{1}{x + iy} = \frac{1}{x + iy} \cdot \frac{x - iy}{x - iy} = \frac{x - iy}{x^2 - i^2 y^2} = \frac{x - iy}{x^2 + y^2}$

$\Rightarrow z^{-1} = \frac{x}{x^2 + y^2} - i \frac{y}{x^2 + y^2}$

- **Division of Two Complex Numbers:** - Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$

$\frac{z_1}{z_2} = \frac{(x_1 + iy_1)}{(x_2 + iy_2)} = \frac{(x_1 + iy_1)(x_2 - iy_2)}{(x_2 + iy_2)(x_2 - iy_2)} = \frac{(x_1 x_2 + y_1 y_2) + i(-x_1 y_2 + x_2 y_1)}{(x_2^2 + y_2^2)}$

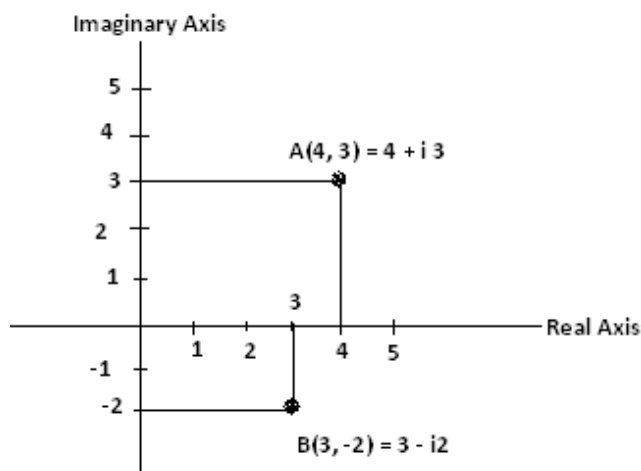
$$\Rightarrow \frac{z_1}{z_2} = \frac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2} + i \frac{x_2y_1 - x_1y_2}{x_2^2 + y_2^2}$$

- **Complex Conjugate:** -Let $z = x + iy$.

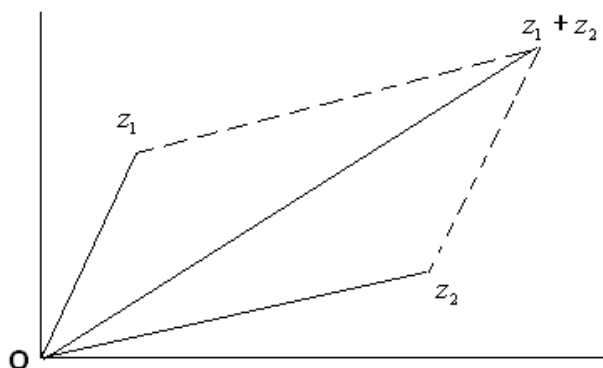
Its complex conjugate $\bar{z} = \overline{x + iy} = x - iy$

So, to get complex conjugate of a complex number just multiply (-1) to its imaginary part.

- **Complex Plane or Argand Plane:** - A plane where complex number shown is called as complex plane or Argand plane. It has two axes. Real axis and Imaginary axis.
- **How To Represent A Complex Number In A Complex Plane:** -

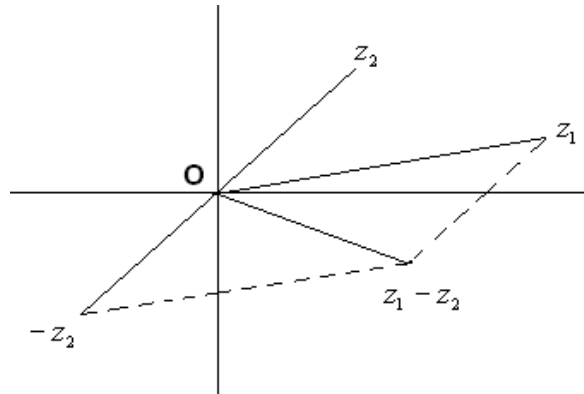


- **How To Represent Sum of Two Complex Numbers In A Complex Plane:** -



Make a line parallel to Oz_1 passes through z_2 and another line parallel to Oz_2 passes through z_1 . They intersect at a point. That point is denoted as $z_1 + z_2$.

• **How To Represent Subtraction of Two Complex Numbers In A Complex Plane: -**



Extend the line z_2O upto $-z_2$ such that distance between Oz_2 and $-z_2O$ are equal. Make a line parallel to Oz_1 passes through $-z_2$ and another line parallel to $-z_2$ passes through z_1 . They intersect at a point. That point is denoted as $z_1 - z_2$.

• **Properties Of Complex Conjugate: -**

Let $z = x + iy$ and $\bar{z} = x - iy$

$$(1) \quad z + \bar{z} = x + iy + x - iy = 2x = 2\text{Re}(z) \Rightarrow \text{Re}(z) = \frac{z + \bar{z}}{2}$$

$$(2) \quad z - \bar{z} = x + iy - x + iy = 2iy = 2i \text{Im}(z) \Rightarrow \text{Im}(z) = \frac{z - \bar{z}}{2i}$$

$$(3) \quad \overline{(\bar{z})} = z$$

$$(4) \quad \overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$$

$$(5) \quad \overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$$

$$(6) \quad \overline{z_1 \times z_2} = \bar{z}_1 \times \bar{z}_2$$

$$(7) \quad \overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}$$

$$(8) \quad \text{If } z = \bar{z} \Rightarrow x + iy = x - iy \Rightarrow 2iy = 0 \Rightarrow y = 0. \text{ Then } z \text{ is a purely real number.}$$

$$(9) \quad \text{If } z = -\bar{z} \Rightarrow x + iy = -x + iy \Rightarrow 2x = 0 \Rightarrow x = 0. \text{ Then } z \text{ is a purely imaginary number.}$$

$$(10) \quad \text{If } R(z) = \frac{P(z)}{Q(z)}, \text{ then } \overline{R(z)} = \frac{P(\bar{z})}{Q(\bar{z})}.$$

$$(11) \quad \text{If } P(z) = a_0 + a_1z + a_2z^2 + \dots + a_nz^n, \text{ then } P(\bar{z}) = \bar{a}_0 + \bar{a}_1(\bar{z}) + \bar{a}_2(\bar{z})^2 + \dots + \bar{a}_n(\bar{z})^n.$$

$$(12) \quad \text{If } Z = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}, \text{ then } \bar{Z} = \begin{vmatrix} \bar{a}_1 & \bar{b}_1 & \bar{c}_1 \\ \bar{a}_2 & \bar{b}_2 & \bar{c}_2 \\ \bar{a}_3 & \bar{b}_3 & \bar{c}_3 \end{vmatrix}.$$

- **Modulus of a Complex Number:** Modulus of a complex number is the distance from the origin to the complex number.

If $z = x + iy$ then modulus of complex number is $|z| = \sqrt{x^2 + y^2}$

- **Properties of modulus of Complex Number:**

- (1) If $|z| = 0$, then $z = 0$.
- (2) $|z| = |-z| = |\bar{z}| = |-\bar{z}|$.
- (3) $Re(z) \leq |z|$ Or $-|z| \leq Re(z) \leq |z|$.
- (4) $Im(z) \leq |z|$ Or $-|z| \leq Im(z) \leq |z|$.
- (5) $|z| \leq |Re(z)| + |Im(z)| \leq \sqrt{2}|z|$.
- (6) $|Re(z)| + |Im(z)| = |z|$, if z is purely real or purely imaginary.
- (7) $|Re(z)| + |Im(z)| = \sqrt{2}|z|$, if $|Re(z)| = |Im(z)|$.
- (8) $|z_1 \times z_2| = |z_1| \times |z_2|$.
- (9) $|z_1 z_2 z_3 \dots z_n| = |z_1| |z_2| |z_3| \dots |z_n|$.
- (10) $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$, if $z_2 \neq 0$.
- (11) $z \times \bar{z} = |z|^2$.
- (12) $|z_1 + z_2| \leq |z_1| + |z_2|$.
- (13) $|z_1 - z_2| \leq |z_1| + |z_2|$.
- (14) $||z_1| - |z_2|| \leq |z_1| + |z_2|$.
- (15) $|z_1 - z_2| \geq ||z_1| - |z_2||$.
- (16) $|z_1 + z_2 + z_3 + \dots + z_n| \leq |z_1| + |z_2| + |z_3| + \dots + |z_n|$.
- (17) $||z_1| - |z_2|| \leq |z_1 + z_2| \leq |z_1| + |z_2|$.
- (18) $|z_1 - z_2| = |z_1| + |z_2|$ holds if $0, z_1, z_2$ are collinear and z_1, z_2 lie on opposite side of origin. i.e. $\arg(z_1) - \arg(z_2) = \pi$ or $\frac{z_1}{z_2} < 0$.
- (19) $|z_1 + z_2| = |z_1| + |z_2|$ holds if $0, z_1, z_2$ are collinear and z_1, z_2 lie on same side of origin. i.e. $\arg(z_1) - \arg(z_2) = 0$ or $\frac{z_1}{z_2} > 0$.
- (20) $|z_1 - z_2| = ||z_1| - |z_2||$ holds if $0, z_1, z_2$ are collinear and z_1, z_2 lie on same side of origin.
- (21) $|z_1 + z_2| = ||z_1| - |z_2||$ holds if $0, z_1, z_2$ are collinear and z_1, z_2 lie on opposite side of origin.
- (22) $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + 2Re(z_1 \bar{z}_2)$.

$$(23) \quad |z_1 - z_2|^2 = |z_1|^2 + |z_2|^2 - 2\operatorname{Re}(z_1 \bar{z}_2).$$

$$(24) \quad |z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2).$$

$$(25) \quad |az_1 + bz_2|^2 + |bz_1 - az_2|^2 = (a^2 + b^2)(|z_1|^2 + |z_2|^2), \text{ where } a, b \in \mathbb{R}.$$

$$(26) \quad \text{If } z_1, z_2 \neq 0, \text{ then } |z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 \Leftrightarrow \frac{z_1}{z_2} \text{ is purely imaginary.}$$

- **Argument of a Complex Number:** It is defined as the angle made by the line joining origin and complex number to the positive x-axis in anticlockwise direction.

If $z = x + iy$ then

$$1. \quad \arg(z) = \theta = \tan^{-1} \frac{y}{x}, \quad \text{if } x > 0.$$

$$2. \quad \arg(z) = \theta = \tan^{-1} \frac{y}{x} + \pi, \quad \text{if } x < 0, y \geq 0.$$

$$3. \quad \arg(z) = \theta = \tan^{-1} \frac{y}{x} - \pi, \quad \text{if } x < 0, y < 0.$$

- **Principal Argument:** Argument of a complex between $(-\pi, \pi]$ is called as principal argument of the complex number.

- **How to find Principal argument:**

(1) Let $z = x + iy$ is the given complex number.

$$(2) \quad \text{Find } \theta = \tan^{-1} \left| \frac{y}{x} \right|.$$

(3) If θ lies in first quadrant, then $\arg(z) = \theta$.

(4) If θ lies in second quadrant, then $\arg(z) = \pi - \theta$.

(5) If θ lies in third quadrant, then $\arg(z) = \theta - \pi$.

(6) If θ lies in fourth quadrant, then $\arg(z) = -\theta$.

(7) If $z = 0$, then $\arg(z)$ is not defined.

- **Properties of Argument of Complex Number:**

$$(1) \quad \text{If } \arg(z) = \theta \text{ then } \arg(\bar{z}) = -\theta.$$

$$(2) \quad \text{If } \arg(z) = \theta \text{ then } \arg\left(\frac{1}{\bar{z}}\right) = -\theta.$$

$$(3) \quad \arg(z_1 z_2) = \arg(z_1) + \arg(z_2).$$

$$(4) \quad \arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2).$$

$$(5) \quad \arg(z^n) = n \arg(z).$$

$$(6) \quad P \arg(z_1 z_2) = P \arg(z_1) + P \arg(z_2) + 2k\pi, \quad k \in \mathbb{Z}.$$

$$(7) \quad P \arg\left(\frac{z_1}{z_2}\right) = P \arg(z_1) - P \arg(z_2) + 2k\pi, \quad k \in \mathbb{Z}.$$

$$(8) \quad P \arg(z^n) = nP \arg(z) + 2k\pi, \quad k \in \mathbb{Z}.$$

- **Polar form of Complex Number:** Let $z = x + iy$ then its polar form is $z = r(\cos\theta + i\sin\theta)$ where $r = |z|$ and $\theta = \arg(z)$.
- **Exponential form of Complex Number:** Let $z = x + iy$ then its exponential form is $z = re^{i\theta} = r(\cos\theta + i\sin\theta)$ where $r = |z|$ and $\theta = \arg(z)$.
- **De-Moivre's Theorem:** For $n \in \mathbb{R}$, $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$.
- $(\cos\alpha + i\sin\alpha)(\cos\beta + i\sin\beta) = \cos(\alpha + \beta) + i\sin(\alpha + \beta)$.
- $\frac{(\cos\alpha + i\sin\alpha)}{(\cos\beta + i\sin\beta)} = \cos(\alpha - \beta) + i\sin(\alpha - \beta)$.
- $\cos\theta + i\sin\theta$ is denoted by $\text{Cis}(\theta)$.

- **Square root of a complex Number:** Let $z = a + ib$ be any complex number, then

$$\sqrt{z} = \sqrt{a + ib} = \begin{cases} \pm \left[\sqrt{\frac{|z|+a}{2}} + i\sqrt{\frac{|z|-a}{2}} \right] & b > 0 \\ \pm \left[\sqrt{\frac{|z|+a}{2}} - i\sqrt{\frac{|z|-a}{2}} \right] & b < 0 \end{cases}$$

- **nth root of unity:** nth root of unity is the solution of $z^n = 1$. It is given by $\text{Cis}\left(\frac{2k\pi}{n}\right)$ for

$$k = 0, 1, 2, 3, \dots, n-1. \text{ Let } \text{Cis}\left(\frac{2\pi}{n}\right) = \alpha$$

$$(1) \quad \text{Then } 1, \alpha, \alpha^2, \alpha^3, \dots, \alpha^{n-1} \text{ are the } n\text{th roots of unity.}$$

$$(2) \quad 1, \alpha, \alpha^2, \alpha^3, \dots, \alpha^{n-1} \text{ are in GP with common ratio } e^{i\frac{2\pi}{n}}.$$

$$(3) \quad 1 + \alpha + \alpha^2 + \alpha^3 + \dots + \alpha^{n-1} = 0 \Rightarrow \sum_{r=0}^{n-1} \alpha^r = 0.$$

$$(4) \quad \frac{1}{x-1} + \frac{1}{x-\alpha} + \frac{1}{x-\alpha^2} + \dots + \frac{1}{x-\alpha^{n-1}} = 0 \Rightarrow \sum_{r=0}^{n-1} \frac{1}{x-\alpha^r} = \frac{nx^{n-1}}{x^n - 1}.$$

$$(5) \quad \sum_{r=0}^{n-1} \cos\left(\frac{2r\pi}{n}\right) = 0 \text{ and } \sum_{r=0}^{n-1} \sin\left(\frac{2r\pi}{n}\right) = 0.$$

$$(6) \quad 1 \times \alpha \times \alpha^2 \times \alpha^3 \times \dots \times \alpha^{n-1} = \prod_{r=0}^{n-1} \alpha^r = \left(\text{Cis}\left(\frac{2\pi}{n}\right)\right)^{\frac{n(n-1)}{2}} = (\text{Cis}(\pi))^{n-1} = (-1)^{n-1} = \begin{cases} 1 & n \text{ even} \\ -1 & n \text{ odd} \end{cases}.$$

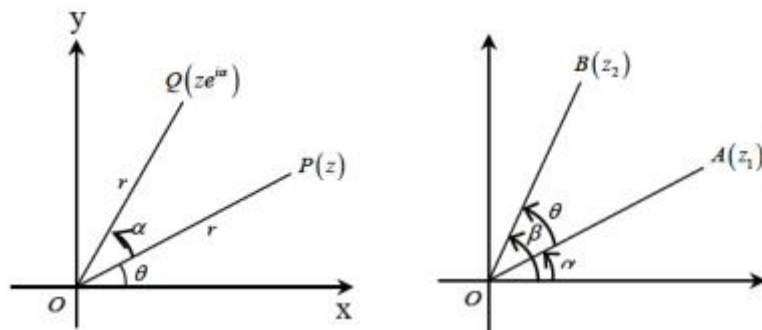
- (7) If $1, \alpha, \alpha^2, \alpha^3, \dots, \alpha^{n-1}$ are the n th roots of unity, then
- $$(1)^m + (\alpha)^m + (\alpha^2)^m + (\alpha^3)^m + \dots + (\alpha^{n-1})^m = \begin{cases} 0 & \text{if } m \text{ is not integral multiple of } n \\ n & \text{if } m \text{ is integral multiple of } n \end{cases}.$$
- (8) If $1, \alpha, \alpha^2, \alpha^3, \dots, \alpha^{n-1}$ are the n th roots of unity, then
- $$(1 + \alpha)(1 + \alpha^2)(1 + \alpha^3) \dots (1 + \alpha^{n-1}) = \prod_{r=1}^{n-1} (1 + \alpha^r) = \begin{cases} 0 & n \text{ even} \\ 1 & n \text{ odd} \end{cases}.$$
- (9) Roots of $z^n = p^n$ are $p, p\alpha, p\alpha^2, p\alpha^3, \dots, p\alpha^{n-1}$ where $1, \alpha, \alpha^2, \alpha^3, \dots, \alpha^{n-1}$ are the n th roots of unity.
- **Cube root of unity:** Cube root of unity is the solution of $z^3 = 1$. It is given by $\text{Cis}\left(\frac{2k\pi}{3}\right)$ for $k = 0, 1, 2$.
- (1) Cube roots are $1, w, w^2$ where $w = \frac{-1+i\sqrt{3}}{2}, w^2 = \frac{-1-i\sqrt{3}}{2}$.
- (2) $w^3 = 1$.
- (3) $1 + w + w^2 = 0, 1 + w = -w^2, 1 + w^2 = -w, w + w^2 = -1$.
- (4) In general $1 + w^m + w^{2m} = \begin{cases} 0 & \text{if } m \text{ is not integral multiple of } 3 \\ 3 & \text{if } m \text{ is integral multiple of } 3 \end{cases}$
- (5) w, w^2 are the roots of $x^2 + x + 1 = 0$.
- (6) $1, w, w^2$ are the vertices of an equilateral triangle inscribed in the circle $|z| = 1$.
- (7) If a, b, c are nonzero real numbers such that $a + b + c = 0 = a^2 + b^2 + c^2$, then $a:b:c = 1:w:w^2$.
- (8) $x^3 + 1 = (x + 1)(x + w)(x + w^2)$.
- (9) $x^3 - 1 = (x - 1)(x - w)(x - w^2)$.
- (10) $(x + y)(x + wy)(x + w^2y) = x^3 + y^3$.
- (11) $(x - y)(x - wy)(x - w^2y) = x^3 - y^3$.
- (12) $(x + wy + w^2z)(x + w^2y + wz) = x^2 + y^2 + z^2 - xy - yz - zx$.
- (13) $(x + y + z)(x + wy + w^2z)(x + w^2y + wz) = x^3 + y^3 + z^3 - 3xyz$.
- (14) $(x + yw)(x - yw)(x + yw^2)(x - yw^2) = x^4 + x^2y^2 + y^4$.
- (15) Cube root of any real number a are $\sqrt[3]{a}, \sqrt[3]{aw}, \sqrt[3]{aw^2}$.
- **n th root of a Complex Number:** Let $z = x + iy$ be any complex number. We have to find its n th root. It is given by $z^{1/n} = r^{1/n} \text{Cis}\left(\frac{\theta + 2k\pi}{n}\right)$ for $k = 0, 1, 2, 3, \dots, n-1$, where $r = |z|$ and $\theta = \arg(z)$.
 - **Geometrical Representation of Multiplying a Complex Number by $e^{i\alpha}$:** Let z be a non-zero complex number.

Let it can be written in polar form $z = r(\cos\theta + i\sin\theta) = re^{i\theta}$, where $r = |z|$ and $\theta = \arg(z)$.

Then $\Rightarrow ze^{i\alpha} = re^{i\theta}e^{i\alpha} = re^{i(\theta+\alpha)}$

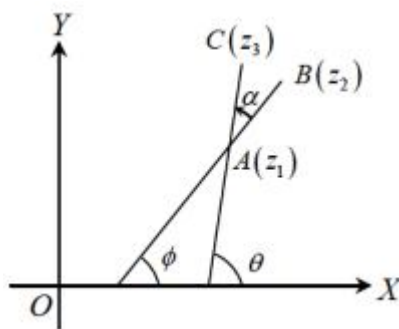
Thus $ze^{i\alpha}$ represents the complex number whose modulus $|ze^{i\alpha}| = r$ and argument $\arg(ze^{i\alpha}) = \theta + \alpha$.

Geometrically $ze^{i\alpha}$ can be obtained by rotating segment joining O and P(z) through an angle α in anticlockwise direction.



Similarly if $A(z_1)$ and $B(z_2)$ are two complex numbers such that angle $\angle AOB = \theta$, then we can write $z_2 = \frac{|z_2|}{|z_1|} z_1 e^{i\theta}$.

• **Rotation Theorem or Coni Formula or Angle between Two Intersecting Lines:**



Let z_1, z_2, z_3 be the affixes of three points A, B, C taken on Argand plane.

Then we have $\overline{AC} = z_3 - z_1$, $\overline{AB} = z_2 - z_1$.

Let $\arg(\overline{AC}) = \arg(z_3 - z_1) = \theta$ and $\arg(\overline{AB}) = \arg(z_2 - z_1) = \phi$

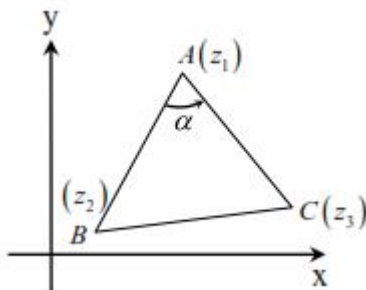
Now, $\angle CAB = \alpha = \theta - \phi = \arg(\overline{AC}) - \arg(\overline{AB})$

$\Rightarrow \alpha = \arg(z_3 - z_1) - \arg(z_2 - z_1)$

$$\Rightarrow \alpha = \arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right).$$

$$\text{Angle between AC and AB} = \arg\left(\frac{\text{affix of } C - \text{affix of } A}{\text{affix of } B - \text{affix of } A}\right).$$

- Interpretation of $\frac{z_3 - z_1}{z_2 - z_1}$.



If z_1, z_2, z_3 are the vertices of a triangle ABC described in the counter clockwise sense then,

$$(1) \quad \arg\left(\frac{\overline{AC}}{AB}\right) = \arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right) = \angle BAC = \alpha.$$

$$(2) \quad \frac{z_3 - z_1}{z_2 - z_1} = \frac{CA}{CB}(\cos\alpha + i\sin\alpha)$$

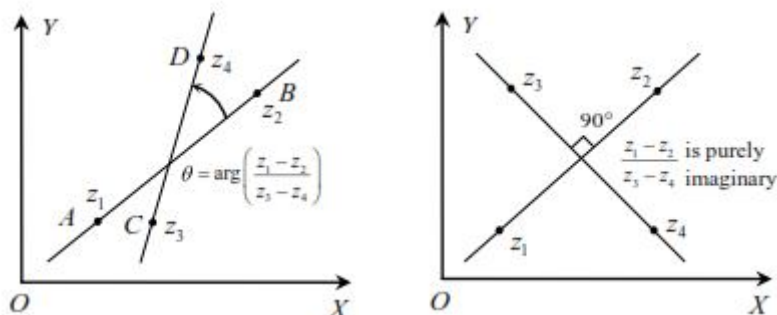
- (3) The points z_1, z_2, z_3 will be collinear if and only if $\alpha = 0$ or π , i.e. if and only if $\frac{z_3 - z_1}{z_2 - z_1}$ is purely real number.

$$(4) \quad \text{The points } z_1, z_2, z_3 \text{ will be collinear if: } \begin{vmatrix} z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \\ z_3 & \bar{z}_3 & 1 \end{vmatrix} = 0.$$

$$(5) \quad \text{The points } z_1, z_2, z_3 \text{ will be collinear if: } \frac{z_3 - z_1}{z_2 - z_1} = \frac{\bar{z}_3 - \bar{z}_1}{\bar{z}_2 - \bar{z}_1}.$$

$$(6) \quad \text{The points } z_1, z_2, z_3 \text{ will be collinear if: } \bar{z}_1(z_2 - z_3) + \bar{z}_2(z_3 - z_1) + \bar{z}_3(z_1 - z_2) = 0.$$

- Interpretation of $\frac{z_1 - z_2}{z_3 - z_4}$.



Let z_1, z_2, z_3, z_4 be four complex numbers.

- (1) Then the line joining z_4 and z_3 is inclined to the line joining z_2 and z_1 at the following angle: $\arg\left(\frac{\overline{CD}}{AB}\right) = \arg\left(\frac{z_4 - z_3}{z_2 - z_1}\right)$.
- (2) The line joining z_4 and z_3 is inclined at 90° to the line joining z_2 and z_1 if: $\arg\left(\frac{z_1 - z_2}{z_3 - z_4}\right) = \pm \frac{\pi}{2}$ i.e. $z_1 - z_2 = \pm ik(z_3 - z_4)$.

• **Different Loci:**

- (1) **Distance Formula:** The distance between two points $P(z_1)$ and $Q(z_2)$ is $PQ = |z_1 - z_2|$.
- (2) **Section Formula:** If $R(z)$ divides the line segment joining $P(z_1)$ and $Q(z_2)$ in the ratio $m : n$ ($m, n > 0$), then:
 - (a) For Internal Division: $z = \frac{mz_2 + nz_1}{m + n}$.
 - (b) For External Division: $z = \frac{mz_2 - nz_1}{m - n}$.
- (3) **Conditions for four points $A(z_1), B(z_2), C(z_3)$ and $D(z_4)$ to represent vertices of:**
 - (a) **Parallelogram** If:
 - (i) $\frac{z_1 + z_3}{2} = \frac{z_2 + z_4}{2} \Rightarrow z_1 + z_3 = z_2 + z_4$.
 - (b) **Rhombus** if:
 - (i) $z_1 + z_3 = z_2 + z_4$.
 - (ii) $|z_4 - z_1| = |z_2 - z_1|$.
 - (iii) $\arg\left(\frac{z_3 - z_1}{z_4 - z_2}\right) = \pm \frac{\pi}{2}$
 - (c) **Rectangle** if:
 - (i) $z_1 + z_3 = z_2 + z_4$.
 - (ii) $|z_3 - z_1| = |z_4 - z_2|$.

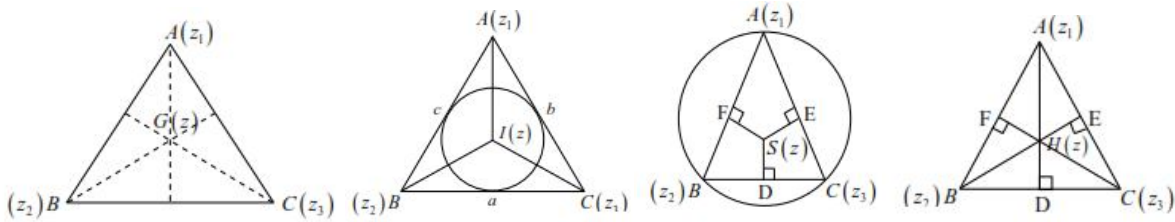
(d) **Square** if:

(i) $z_1 + z_3 = z_2 + z_4$.

(ii) $|z_4 - z_1| = |z_2 - z_1|$.

(iii) $|z_3 - z_1| = |z_4 - z_2|$.

(4) **Centroid, Incentre, Orthocentre and Circumcentre of a Triangle:** Let ABC be a triangle with vertices $A(z_1), B(z_2), C(z_3)$,



(a) **Centroid G(z):** $z = \frac{z_1 + z_2 + z_3}{3}$.

(b) **Incentre I(z):** $z = \frac{az_1 + bz_2 + cz_3}{a + b + c}$.

(c) **Circumcentre S(z):** $z = \frac{(z_2 - z_3)|z_1|^2 + (z_3 - z_1)|z_2|^2 + (z_1 - z_2)|z_3|^2}{(z_2 - z_3)z_1 + (z_3 - z_1)z_2 + (z_1 - z_2)z_3}$.

$$\text{Or } \Rightarrow z = \frac{\begin{vmatrix} |z_1|^2 & z_1 & 1 \\ |z_2|^2 & z_2 & 1 \\ |z_3|^2 & z_3 & 1 \end{vmatrix}}{\begin{vmatrix} z_1 & z_1 & 1 \\ z_2 & z_2 & 1 \\ z_3 & z_3 & 1 \end{vmatrix}}.$$

$$\text{Or } z = \frac{z_1(\sin 2A) + z_2(\sin 2B) + z_3(\sin 2C)}{\sin 2A + \sin 2B + \sin 2C}.$$

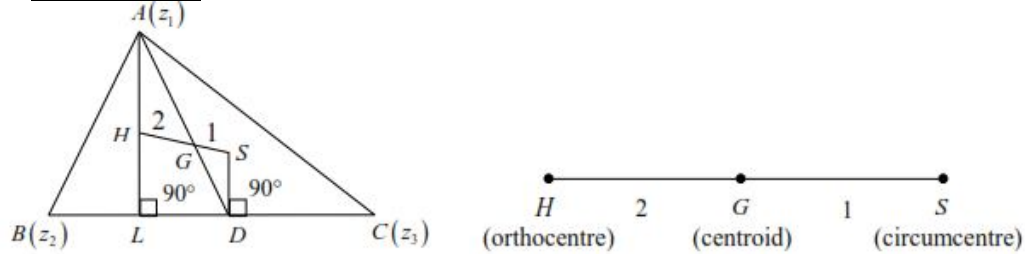
(d) **Orthocentre H(z):** $z = \frac{z_1(\tan A) + z_2(\tan B) + z_3(\tan C)}{\tan A + \tan B + \tan C}$

$$\text{Or } \Rightarrow z = \frac{z_1(a \sec A) + z_2(b \sec B) + z_3(c \sec C)}{a \sec A + b \sec B + c \sec C}$$

$$\text{Or } \Rightarrow z = \frac{\begin{vmatrix} |z_1|^2 & z_1 & 1 \\ |z_2|^2 & z_2 & 1 \\ |z_3|^2 & z_3 & 1 \end{vmatrix}}{\begin{vmatrix} z_1 & z_1 & 1 \\ z_2 & z_2 & 1 \\ z_3 & z_3 & 1 \end{vmatrix}}.$$

$$z = \frac{\bar{z}_1(z_2 - z_3)(z_2 + z_3 - z_1) + \bar{z}_2(z_3 - z_1)(z_3 + z_1 - z_2) + \bar{z}_3(z_1 - z_2)(z_1 + z_2 - z_3)}{\bar{z}_1(z_2 - z_3) + \bar{z}_2(z_3 - z_1) + \bar{z}_3(z_1 - z_2)}.$$

(5) **Euler's Line:**



The centroid G of a triangle lies on the segment joining the orthocentre H and circumcentre S of the triangle. G divides the join of H and S in the ratio of 2:1. This line is called as Euler's line.

(6) **Area of a triangle:** Let ABC be a triangle with vertices $A(z_1), B(z_2), C(z_3)$, then its

$$\text{area is given by } \Delta ABC = \frac{1}{4} \left| \begin{vmatrix} z_1 & \bar{z}_1 & 1 \\ i z_2 & \bar{z}_2 & 1 \\ z_3 & \bar{z}_3 & 1 \end{vmatrix} \right| = \frac{1}{4} \left| \begin{vmatrix} z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \\ z_3 & \bar{z}_3 & 1 \end{vmatrix} \right|.$$

$$\Rightarrow \text{Area} = \frac{\bar{z}_1(z_2 - z_3) + \bar{z}_2(z_3 - z_1) + \bar{z}_3(z_1 - z_2)}{4}.$$

(7) **Condition for Triangle to be Equilateral:** Let ABC be a triangle with vertices $A(z_1), B(z_2), C(z_3)$, then ABC is an equilateral triangle if and only if :

$$\frac{1}{z_1 - z_2} + \frac{1}{z_2 - z_3} + \frac{1}{z_3 - z_1} = 0$$

$$\Rightarrow z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$$

$$\Rightarrow \begin{vmatrix} 1 & z_2 & z_3 \\ 1 & z_3 & z_1 \\ 1 & z_1 & z_2 \end{vmatrix} = 0.$$

(8) **Equation of Straight Line:**

(a) **Non-Parametric Form:** Equation of a line joining the points $A(z_1), B(z_2)$ is

$$\begin{vmatrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix} = 0$$

$$\Rightarrow \frac{z - z_1}{z_2 - z_1} = \frac{\bar{z} - \bar{z}_1}{\bar{z}_2 - \bar{z}_1}$$

$$\Rightarrow z(\overline{z_1 - z_2}) - \overline{z}(z_1 - z_2) + z_1 \overline{z_2} - z_2 \overline{z_1} = 0.$$

- (b) **Parametric Form:** Equation of a line joining the points $A(z_1), B(z_2)$ is $z = tz_1 + (1-t)z_2$.

- (c) **General Equation of Straight Line:** The general equation of straight line is $\overline{a}z + a\overline{z} + b = 0$, where a is a complex number and b is a real number.

- (d) **Equation of the perpendicular bisector:** If $A(z_1), B(z_2)$ are two fixed points and $R(z)$ is moving point such that it is always at equal distance from $A(z_1)$ and $B(z_2)$ then locus of $R(z)$ is the perpendicular bisector of PQ i.e. $PR = QR$

$$\Rightarrow |z - z_1| = |z - z_2|.$$

$$\Rightarrow |z - z_1|^2 = |z - z_2|^2.$$

$$\Rightarrow z(\overline{z_1 - z_2}) + \overline{z}(z_1 - z_2) = |z_1|^2 - |z_2|^2.$$

- (9) **Complex Slope of a line:** If $A(z_1)$ and $B(z_2)$ are two points in the complex plane, then complex slope of AB is defined to be $\mu = \frac{z_1 - z_2}{\overline{z_1} - \overline{z_2}}$.

The complex slope of the line $\overline{a}z + a\overline{z} + b = 0$ is given by $-\frac{a}{\overline{a}}$.

Let two lines with complex slopes μ_1 and μ_2 , then:

- (a) Two lines are parallel if $\mu_1 = \mu_2$.

- (b) Two lines are perpendicular if $\mu_1 + \mu_2 = 0$.

- (c) Angle between two lines $\tan \theta = \left| \frac{i(\mu_2 - \mu_1)}{\mu_2 + \mu_1} \right|$.

- (10) **Equation of Circle:**

- (a) An equation of circle with centre z_0 and radius r is $|z - z_0| = r$.

$$\Rightarrow z\overline{z} - z_0\overline{z} - \overline{z_0}z + z_0\overline{z_0} - r^2 = 0.$$

- (b) **General Equation of Circle:** General equation of a circle is $z\overline{z} + a\overline{z} + \overline{a}z + b = 0$.

- (c) **Diametric Form of a circle:** An equation of a circle one of whose diameter is the segment joining $A(z_1)$ and $B(z_2)$ is: $Re\left(\frac{z - z_1}{z - z_2}\right) = 0$ or

$$\Rightarrow (z - z_1)(\overline{z} - \overline{z_2}) + (\overline{z} - \overline{z_1})(z - z_2) = 0.$$

- (d) **Equation of Circle Passing through two points:** Equation of circle passes through two points $A(z_1)$ and $B(z_2)$ is:

$$(z - z_1)(\overline{z} - \overline{z_2}) + (\overline{z} - \overline{z_1})(z - z_2) + k \begin{vmatrix} z & \overline{z} & 1 \\ z_1 & \overline{z_1} & 1 \\ z_2 & \overline{z_2} & 1 \end{vmatrix} = 0, \text{ where } k \text{ is a real parameter.}$$

- (e) **Equation of three non-collinear points:** Equation of circle passes through three non-collinear points $A(z_1)$, $B(z_2)$ and $C(z_3)$ is:

$$\frac{(z - z_1)(z_3 - z_2)}{(z - z_2)(z_3 - z_1)} = \frac{(\bar{z} - \bar{z}_1)(\bar{z}_3 - \bar{z}_2)}{(\bar{z} - \bar{z}_2)(\bar{z}_3 - \bar{z}_1)}.$$

- (f) **Condition for four points to be concyclic:** Four points z_1, z_2, z_3, z_4 will lie on the same circle if:

$$\Rightarrow \frac{(z_4 - z_1)(z_3 - z_2)}{(z_4 - z_2)(z_3 - z_1)} \text{ is purely real but } \frac{z_4 - z_1}{z_4 - z_2} \text{ is non-real.}$$

$$\Rightarrow \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)} \text{ is purely real but } \frac{z_1 - z_3}{z_1 - z_4} \text{ is non-real.}$$

- (11) If α is a real number, then $\arg(z) = \alpha$ represents a ray starting at the origin and making an angle α with real axis.
- (12) If α is a real number and z_0 is a fixed point, then $\arg(z - z_0) = \alpha$ represents a ray starting at z_0 and making an angle α with real axis.
- (13) Locus of z satisfying $|z - z_0| = r$ is a circle with centre z_0 and radius r .
- (14) $|z - z_0| < r$ represents the region inside the circle with centre z_0 and radius r .
- (15) $|z - z_0| > r$ represents the region outside the circle with centre z_0 and radius r .
- (16) $a < |z - z_0| < b$ represents the region lying between the two concentric circles $|z - z_0| = a$ and $|z - z_0| = b$.
- (17) Locus of z satisfying $|z - z_1| = |z - z_2|$ is the perpendicular bisector of the line segment joining z_1 and z_2 .
- (18) If z is such that $|z - z_1| = |z - z_2| = |z - z_3|$, then z is the circumcentre of triangle formed by the vertices z_1, z_2 and z_3 .
- (19) If $\frac{|z - z_1|}{|z - z_2|} = k$, ($k \neq 1, > 0$), then the locus of z is Apollonius circle with $\frac{kz_2 + z_1}{k + 1}$, $\frac{kz_2 - z_1}{k - 1}$ as diametrical end points and with radius $\frac{k|z_1 - z_2|}{|k^2 - 1|}$.
- (20) Locus of z satisfying $Re\left(\frac{z - z_1}{z - z_2}\right) = 0$ is a circle with z_1 and z_2 as end points of diameters.
- (21) Locus of z satisfying $\frac{z - z_1}{z - z_2} = \lambda i$ is a circle with z_1 and z_2 as end points of diameters.
- (22) Locus of z satisfying $Arg\left(\frac{z - z_1}{z - z_2}\right) = \pm \frac{\pi}{2}$ is a circle with z_1 and z_2 as end points of diameters.

- (23) Locus of z satisfying $\text{Arg}\left(\frac{z-z_1}{z-z_2}\right) = \frac{\pi}{2}$ is a semicircle with z_1 and z_2 as end points of diameters. (Locus exclude the points z_1 and z_2).
- (24) Locus of z satisfying $\text{Arg}\left(\frac{z-z_1}{z-z_2}\right) = \theta$ is a major arc or minor arc of a circle with z_1 and z_2 as end points of a chord of a circle. (Locus exclude the points z_1 and z_2).
- (25) Locus of z satisfying $\text{Im}\left(\frac{z-z_1}{z-z_2}\right) = 0$ is a line passing through z_1 and z_2 .
- (26) Locus of z satisfying $\text{Arg}\left(\frac{z-z_1}{z-z_2}\right) = 0$ is a line passing through z_1 and z_2 . (Locus exclude the points z_1 and z_2).
- (27) Locus of z satisfying $\text{Arg}\left(\frac{z-z_1}{z-z_2}\right) = \pi$ is a line passing through z_1 and z_2 .
- (28) Locus of z satisfying $\text{Arg}(z-z_1) = \theta$ is a ray with vertex at z_1 and making angle θ with +ve direction of real axis. (Locus exclude the point z_1).
- (29) Locus of z satisfying $|z-z_1| + |z-z_2| = 2a$, when $2a > |z_1 - z_2|$ is an ellipse with z_1 and z_2 as foci and $2a$ as length of major axis and eccentricity $e = \frac{|z_1 - z_2|}{2a}$.
- (30) Locus of z satisfying $|z-z_1| + |z-z_2| = 2a$, when $2a = |z_1 - z_2|$ is a line segment joining z_1 and z_2 .
- (31) Locus of z satisfying $|z-z_1| + |z-z_2| = 2a$, when $2a < |z_1 - z_2|$, represents no any point in argand plane.
- (32) Locus of z satisfying $||z-z_1| - |z-z_2|| = 2a$, when $2a < |z_1 - z_2|$ is a hyperbola with z_1 and z_2 as foci and eccentricity $e = \frac{|z_1 - z_2|}{2a}$.
- (33) Locus of z satisfying $||z-z_1| - |z-z_2|| = 2a$, when $2a = |z_1 - z_2|$ is a line segment joining z_1 and z_2 . (Locus exclude the points z_1 and z_2).
- (34) Locus of z satisfying $||z-z_1| - |z-z_2|| = 2a$, when $2a > |z_1 - z_2|$, represents no any point in argand plane.
- (35) Locus of z satisfying $|z-z_1|^2 + |z-z_2|^2 = k$ is a circle if $k \geq \frac{|z_1 - z_2|^2}{2}$.
- (36) If l, m, n are three real numbers not all zero simultaneously such that $lz_1 + mz_2 + nz_3 = 0$ and $l + m + n = 0$, then the points z_1, z_2 and z_3 are collinear.
- (37) The equation of line joining z_1 and z_2 is $z = z_1 + t(z_2 - z_1)$, t is a parameter.

- (38) General equation of line in complex plane is $\bar{\alpha}z + \alpha\bar{z} + \beta = 0$ where α is non zero complex constants and β is real constant.
- (39) Equation of line parallel to $\bar{\alpha}z + \alpha\bar{z} + \beta = 0$ is $\bar{\alpha}z + \alpha\bar{z} + \gamma = 0$ where β and γ are real constants.
- (40) Equation of line perpendicular to $\bar{\alpha}z + \alpha\bar{z} + \beta = 0$ is $\bar{\alpha}z - \alpha\bar{z} + i\gamma = 0$ where β and γ are real constants.
- (41) General form of equation of circle is $z\bar{z} + \bar{\alpha}z + \alpha\bar{z} + \beta = 0$ where centre of the circle is $-\alpha$ and radius is $\sqrt{\alpha\bar{\alpha} - \beta}$.
- (42) The equation of a circle through 3 non collinear points z_1, z_2, z_3 can be taken as $\frac{z - z_2}{z - z_1} \times \frac{z_3 - z_1}{z_3 - z_2}$ is real $\Rightarrow \frac{z - z_2}{z - z_1} \times \frac{z_3 - z_1}{z_3 - z_2} = \frac{\bar{z} - \bar{z}_2}{\bar{z} - \bar{z}_1} \times \frac{\bar{z}_3 - \bar{z}_1}{\bar{z}_3 - \bar{z}_2}$.
- (43) If z_1, z_2, z_3, z_4 are vertices of a quadrilateral then, it is a cyclic quadrilateral if and only if $|z_1 - z_3| \times |z_2 - z_4| \leq |z_1 - z_2| \times |z_3 - z_4| + |z_1 - z_4| \times |z_2 - z_3|$.
- (44) If $z_1, z_2, z_3, \dots, z_n$ are vertices of n sides regular polygon with z_0 is its Centroid then:
- (a) $z_1 + z_2 + z_3 + \dots + z_n = nz_0$.
- (b) $z_1^2 + z_2^2 + z_3^2 + \dots + z_n^2 = nz_0^2$.
- (45) If z_1, z_2, z_3 be the vertices of equilateral triangle, then:
- (a) $(z_1 - z_2)^2 + (z_2 - z_3)^2 + (z_3 - z_1)^2 = 0$.
- (b) $z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$.
- (c) $\frac{1}{z_1 - z_2} + \frac{1}{z_2 - z_3} + \frac{1}{z_3 - z_1} = 0$.
- (d) $(z_1 - z_2)^2 = (z_2 - z_3)(z_3 - z_1)$.
- **Logarithm of a Complex Number:** $\log_e z = \log_e |z| + i(2n\pi + \arg(z))$.
